NAI-501 FUNDAMENTALS OF DATA ANALYTICS

Assignment: 3

Unit – 3 Data Analysis

1. What is hypothesis testing? Explain why hypothesis testing is used in data analysis. Write the main steps of hypothesis testing in simple words.
2. What is a null hypothesis (H_0) and what is an alternative hypothesis (H_1)? Explain both and give three simple examples of H_0 and H_1 from daily life.
3. What do you mean by Type I error and Type II error in hypothesis testing? Explain both errors with easy, real-life examples.
4. What is level of significance (α) in hypothesis testing? Why is the 5% level of significance commonly used? Explain in simple language.
5. What is a p-value? How is p-value used to decide whether to reject or not reject the null hypothesis? Explain with a small, simple example.
6. What is a Z-test? When do we use the Z-test in hypothesis testing? Write the main conditions and give one simple example where Z-test can be applied.
7. What is a t-test? When is the t-test used instead of the Z-test? Explain with a simple example, such as comparing the average marks of students in a small class.
8. What is regression analysis? Explain the basic idea of regression as finding a relationship between one dependent variable and one independent variable, with a simple example (for example, study hours and exam marks).